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Graph Limits

It is to note that in the below typesetting, a colon is used in set definitions instead of a vertical bar to separate objects and conditions. Further italics are created by the "\textit{}" command instead of the "\emph{}" command (the latter should be used as it is better for handling logical markup). Mathmode is achieved by $ instead of the pairings \[and \]. There are likely other mannerisms present that do not match the original typesetting.

1.1 Nov. 4 Notes

In chapter 1 we defined a graph (G, R) as a structure where G is a set of points, and R as some relation on G. Using ideas defined in chapter 2 we can begin discussing limits of graphs.

Consider the family of a graph

$$(G_n, R_n)_{n\in\mathbb{N}} \xrightarrow{\mathscr{U}} (G, R)$$

where (G, R) is the $\mathscr{U}$-limit of the family $(G_n, R_n)_{n\in\mathbb{N}}$.

Suppose we are given a family of graphs

$$(G_1, R_1), (G_2, R_2), \ldots$$

an an ultrafilter $\mathscr{U}$ on $\mathbb{N}$, we define the $\mathscr{U}$-limit (G, R) of the family $(G_n, R_n)_{n\in\mathbb{N}}$ as follows.

Consider the Cartesian product $\prod_{n\in\mathbb{N}} G_n$ which consists of all the sequences $(a_n)_{n\in\mathbb{N}}$, where $a_n \in G$ for all $n \in \mathbb{N}$, this will be G.

Now for R, suppose $a, b \in \prod_{n\in\mathbb{N}} G_n$, i.e.,

$$a = (a_1, a_2, \ldots)$$
$$b = (b_1, b_2, \ldots)$$

we define

$$a\ R\ b = \{n \in \mathbb{N} : a_n\ R_n\ b_n\} \in \mathscr{U}$$
$$a\ \not R\ b = \{n \in \mathbb{N} : a_n\ \not R_n\ b_n\} \in \mathscr{U}.$$

This can be seen as finding whether almost every element in a is related to, or not related to, every element in b pairwise.

1.2 Nov. 11 Notes

Consider we are given a sequence of graphs

$$\mathscr{G}_1 = (G_1, R_1), \mathscr{G}_2 = (G_2, R_2), \dots$$

Q1: Do these graphs converge to a graph-limit?

Q2: Which properties of $\mathscr{G}_1, \mathscr{G}_2, \dots$ "transfer" to the graph-limit and which do not?

Fix a non-principle ultrafilter $\mathscr{U}$ on the indexing set $\mathbb{N} = \{1, 2, \dots\}$ we will show there is a graph $\mathscr{G}$ such that

$$\mathscr{G}_n = (G_n, R_n)_{n\in\mathbb{N}} \xrightarrow{\mathscr{U}} (G, R) = \mathscr{G}$$

Iovino said that the below section should be at the start of the chapter.

Remark, given any sets $A_1, A_2, \dots$ we define the Cartesian product

$$\prod_{n\in\mathbb{N}} A_n = \{(x_n)_{n\in\mathbb{N}} : x_n \in A_n\}$$

If $x = (x_n)_{n\in\mathbb{N}}$ is a sequence in $\prod_{n\in\mathbb{N}} A_n$ this is called the *n-th coordinate* of x.

The function

$$P_n : \prod_{j\in\mathbb{N}} A_j \to A_n$$
$$x \mapsto x_n$$

End of section

Now constructing the $\mathscr{U}$-limit of $\mathscr{G}_1, \mathscr{G}_2, \dots$

First we let $G = \prod_{n\in\mathscr{N}} G_n$. Now fix any $x, y \in G$, then

$$x = (x_1, x_2, \dots)$$
$$y = (y_1, y_2, \dots).$$

Define a relation on G such that

$$x \sim y \iff \{n : x_n = y_n\} \in \mathscr{U}$$

Claim: $\sim$ is an equivalence relation, i.e., the relation is reflexive, symmetric, and transitive.

Proof. We will show the relation $\sim$ is reflexive, symmetric, and transative. Let $x = (x_n)_{n\in\mathbb{N}}$, $y = (y_n)_{n\in\mathbb{N}}$, and $z = (z_n)_{n\in\mathbb{N}}$ with $x \sim y$ and $y \sim z$.

(1) The relation $\sim$ is reflexive since

$$x \sim x \iff \{n : x_n = x_n\} = \mathbb{N} \in \mathscr{U}$$

The entire indexing set $\mathbb{N}$ is $\mathscr{U}$-large by property 1 of a filter.

(2) The relation $\sim$ is symmetric since

$$x \sim y \iff \{n : x_n = y_n\} = \{n : y_n = x_n\} \iff y \sim x$$

therefore $x \sim y \iff y \sim x$.

(3) The relation $\sim$ is transitive since

$$x \sim y \text{ and } y \sim z \iff \{n : x_n = y_n\} \cap \{n : y_n = z_n\}$$

and

$$x \sim z \iff \{n : x_n = y_n \text{ or } y_n = z_n\} \leftarrow \text{check this}$$

we observe

$$\{n : x_n = y_n \text{ or } y_n = z_n\} \supseteq \{n : x_n = y_n\} \cap \{n : y_n = z_n\}.$$

Filters are closed under supersets, hence $x \sim z$ holds.

$\square$

If $x, y \in G$, we regard x and y as equal if $x \sim y$.
Second, for $x, y \in G$ define $x \; R \; y$ if and only if $x_n \; R_n \; y_n$ almost everywhere (a.e.).

$$x \; R \; y = \{n : x_n \; R_n \; y_n\} \in \mathscr{U}$$
$$x \not R \; y = \{n : x_n \not R_n \; y_n\} \in \mathscr{U}$$

Thusly we have defined the graph-limit $\mathscr{G} = (G, R)$.

Definition 1.1. The graph $\mathscr{G}_{\mathscr{U}} = (G, R)$ is called the $\mathscr{U}$-*Ultraproduct* or the $\mathscr{U}$-*limit* of $\mathscr{G}_1, \mathscr{G}_2, \ldots$.

Remark: The preceeding construction works just as well if we replace $\mathbb{N}$ by any infinite set I or if we replace $\mathscr{G}_n$ by relational structure $(G_i, R_i)_{i \in I}$.
Notation:

$$\mathscr{G}_n \xrightarrow{\mathscr{U}} \mathscr{G}_{\mathscr{U}}$$

or

$$\mathscr{G}_{\mathscr{U}} = \mathscr{U}\text{-}\lim \mathscr{G}_n$$

or

$$\mathscr{G}_{\mathscr{U}} = \prod_{n \in \mathbb{N}} \mathscr{G}_n \setminus \mathscr{U}. \leftarrow \text{check this}$$

1.2.1 Transfer Theorems.

Definition 1.2. *First order statements* are made up of Logical Connectives and Quantifiers

Logical Connectives: $\neg$ (NOT), $\wedge$ (AND), $\vee$ (OR)

Quantifiers: $\forall$ (universal quantification), $\exists$ (existential quantification)

Let $\mathscr{U}$ be an ultrafilter on I, an indexed set. If $A, B \subseteq I$, then

(1) $A \cap B \in \mathscr{U}$ if and only if $A \in \mathscr{U}$ and $B \in \mathscr{U}$

(2) $A \cup B \in \mathscr{U}$ if and only if $A \in \mathscr{U}$ or $B \in \mathscr{U}$

(3) $A^{\mathsf{c}} \in \mathscr{U}$ if and only if $A \notin \mathscr{U}$

Proof. Let $\mathscr{U}$ be an ultrafilter on I, an indexed set and $A, B \subseteq I$.

(1) Assume $A \cap B \in \mathscr{U}$, then $A \supseteq A \cap B$ and $B \supseteq A \cap B$ so $A \in \mathscr{U}$. Since filters are closed under supersets, thus $A \in \mathscr{U}$ and $B \in \mathscr{U}$. Conversely, assume $A \in \mathscr{U}$ and $B \in \mathscr{U}$, then $A \cap B \in \mathscr{U}$ by the intersection property of filters.

(2) Assume $A \cup B \in \mathscr{U}$ and $A^{\mathsf{c}}, B^{\mathsf{c}} \in \mathscr{U}$, then by closure of intersection $A^{\mathsf{c}} \cap B^{\mathsf{c}} \in \mathscr{U}$, by De Morgan's Laws† $A^{\mathsf{c}} \cap B^{\mathsf{c}} = (A \cup B)^{\mathsf{c}}$, this is a contraction since reasoning, hence $A \in \mathscr{U}$ or $B \in \mathscr{U}$. Conversly, assume $A \in \mathscr{U}$ or $B \in \mathscr{U}$, since $A \cup B \supseteq A \in \mathscr{U}$ or $A \cup B \supseteq B \in \mathscr{U}$, by the closure under supersets it follows that $A \cup B \in \mathscr{U}$.

†: theorem required

(3) This holds by the definition of an ultrafilter. □

Corollary 1.3. *Consider a family of graphs*

$$\mathscr{G}_1 = (G_1, R_1), \mathscr{G}_2 = (G_2, R_2), \ldots$$

Let $\mathscr{G}$ be the $\mathscr{U}$-limit of $\mathscr{G}_1, \mathscr{G}_2, \ldots$ if $p(x), q(x)$ are properties [should we define what a property is?] of graphs, we say

$$p(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : p(x) \textit{ true}\} \in \mathscr{U}$$

check below

$$p(x) \wedge q(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : p(x) \wedge q(x) \textit{ true}\} \in \mathscr{U}$$
$$p(x) \vee q(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : p(x) \vee q(x) \textit{ true}\} \in \mathscr{U}$$
$$\neg p(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : \neg p(x) \textit{ true}\} \in \mathscr{U}$$
$$\exists p(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : \exists p(x) \textit{ true}\} \in \mathscr{U}$$
$$\forall p(x) \textit{ holds in } \mathscr{G} \textit{ if and only if } \{x \in \mathscr{G} : \forall p(x) \textit{ true}\} \in \mathscr{U}$$